

No. of Pages	3
No. of Questions	4
Total Marks	25
Time: 75 minutes	

Department of Computer Science and Engineering
Midterm Examination Fall 2025
CSE 110: Programming Language I

- ✓ Write your Set Number on top of the answer script in LARGE FONT.
- ✓ Answer all questions. Use back part of the answer script for rough work.
- ✓ Answer Question 1 & 2 at the beginning part of the answer script.
- ✓ Figure in bracket [] next to each question indicates marks for that question.
- ✓ At the end of exam, put question paper inside answer script and return both.
- ✓ Understanding the question is part of the exam, please do not ask questions



Section: _____ ID: _____ Name in CAPITAL: _____

Question 1 [CO3] [6 Points]
[Answer on the answer-script]

Draw a flowchart that takes a candidate's years of experience (exp), skill score (sScore), and test score (tScore) as inputs from the user and prints whether the candidate is "Rejected," "Shortlisted," or "Waitlisted" based on the following criteria.

1. Candidates with experience of less than 2 years are Rejected.
2. Candidates with experience of 2 years or more are further evaluated based on a score calculated using the formula:
 - A. If $2 \leq \text{exp} \leq 4$:

$$\text{score} = \text{sScore} * 0.6 + \text{tScore} * 0.4$$
 - B. Otherwise:

$$\text{score} = \text{sScore} * 0.6 + \text{tScore} * 0.4 + 5$$
3. Candidates with score ≥ 80 are Shortlisted. Otherwise, they are Waitlisted.

Question 2 [CO1] [10 Points]
[Answer on the answer-script]

Harshad number is a number that is divisible by the sum of its digits. 18 is a Harshad number because $1+8=9$, and 18 is divisible by 9. 15 is not a Harshad number because $1+5=6$, and 15 is not divisible by 6.

Write a Java program that takes two numbers, a and b, as input from the user and prints the total number of Harshad numbers between a and b (inclusive). For simplicity, assume that both numbers are positive and $b > a$

Sample Input	Sample Output	Explanation
a = 1 b = 50	23	Between 1 and 50, the Harshad numbers are: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 18, 20, 21, 24, 27, 30, 36, 40, 42, 45, 48 and 50. There are 23 such numbers in total.
a = 51 b = 100	10	Between 51 and 100, the Harshad numbers are: 54, 60, 63, 70, 72, 80, 81, 84, 90, and 100. There are 10 such numbers in total.

Question 3 [CO1] [5 Points]
[Answer on the question paper]

Assume Brac University has 12 floors and the escalators and lift are currently not working. As a BracU student, you now have to cover so many steps for getting from one floor to another, you now have to cover so many steps for getting from one floor to another and then from one block to another. Tiring right? Well, here's a code to help you calculate the steps you take. Suppose you are on the 12th floor, H block. The floor and block you want to go to from here are taken as user inputs (both integers where A - H block = 1 - 8).

Step difference between two floors = 10 steps
Step difference between two blocks = 20 steps

There are 6 errors in this code. Figure out the errors and write the corrected lines in the box below. One correction has been done for you. Do not entirely omit a code part, just correct the given code.

```

1 import java.util.Scanner;
2 public class ErrorCorrection{
3     public static main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int steps = 0;
6         System.out.println("Which floor do you want to go?");
7         int floor_to_go = sc.nextInt();
8         System.out.println("Which block do you want to go?");
9         int block_to_go = sc.nextInt();
10        for(int i = 12; i >= floor_to_go; i--)
11        {
12            steps += 10;
13        }
14        for(int i = 8; i > block_to_go; i++)
15        {
16            steps -= 20;
17        }
18        System.out.println("Steps covered:" , steps);
19    }
20 }

```

Write your corrections in the table given below. For better understanding, error correction is shown.

Line Number	Fix
Line 3	public static void main(String [] args){
Line 1	import java.util. Scanner ;
Line 14	for (int i=8; i < block_to_go; i++)
Line 16	Steps += 20;
Line 10	for(int i=12; i > floor_to_go; i--)
Line 18	System.out.println ("Steps covered:" + steps);

Question 4 [CO1] [4 Points]
[Answer on the question paper]

Illustrate the outputs of the following statements. Provide the answer script to verify your outputs. Your answer will be marked correct only if you provide the correct outputs without the workings. All of the outputs must be in the question script.

```
1 public class TracingA(  
2     public static void main(String[] args){  
3         int sum = 0, r = 0, i = 8, j = 5;  
4         while (i < 100) {  
5             r = (i%2) + 1;  
6             if (r%2 == 1){  
7                 j = (++i) + j - r + 2;  
8                 continue;  
9             }  
10            sum = i % r + (i * r) - j + 32;  
11            System.out.println(sum);  
12            if (i > 13) {  
13                sum += i/3 + r + 12;  
14                break;  
15            }  
16            else{  
17                i += 4;  
18            }  
19        }  
20        System.out.println(sum);  
21    }  
22 }
```

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No washroom breaks

Section: 15 ID: 1000068012 Name in CAPITAL: SHAH MUSTAGHFIR BILLAH NABIL

Question 1 [CO3] [6 Points]

[Answer on the answer-script]

Draw a flowchart that takes a candidate's years of experience (exp), skill score (sScore), and test score (tScore) as inputs from the user and prints whether the candidate is "Rejected," "Shortlisted," or "Waitlisted" based on the following criteria.

- Candidates with experience of less than 2 years are Rejected.
- Candidates with experience of 2 years or more are further evaluated based on a score calculated using the formula:

A. If $2 \leq \text{exp} \leq 4$:

$$\text{score} = \text{sScore} * 0.6 + \text{tScore} * 0.4$$

B. Otherwise:

$$\text{score} = \text{sScore} * 0.6 + \text{tScore} * 0.4 + 5$$

- Candidates with score ≥ 80 are Shortlisted. Otherwise, they are Waitlisted.

Question 2 [CO1] [10 Points]

[Answer on the answer-script]

Harshad number is a number that is divisible by the sum of its digits. 18 is a Harshad number because $1+8=9$, and 18 is divisible by 9. 15 is not a Harshad number because $1+5=6$, and 15 is not divisible by 6.

Write a Java program that takes two numbers, a and b, as input from the user and prints the total number of Harshad numbers between a and b (inclusive). For simplicity, assume that both numbers are positive and $b > a$

Sample Input	Sample Output	Explanation
a = 1 b = 50	23	Between 1 and 50, the Harshad numbers are: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 18, 20, 21, 24, 27, 30, 36, 40, 42, 45, 48 and 50. There are 23 such numbers in total.
a = 51 b = 100	10	Between 51 and 100, the Harshad numbers are: 54, 60, 63, 70, 72, 80, 81, 84, 90, and 100. There are 10 such numbers in total.

Question 3 [CO1] [5 Points][Answer on the question paper]

Assume Brac University has 12 floors and the escalators and lift are currently not working. As a BracU student, you now have to cover so many steps for getting from one floor to another and then from one block to another. Tiring right? Well, here's a code to help you calculate the steps you take. Suppose you are on the 1st floor, A block. The floor and block you want to go to from here are taken as user inputs (both integers where A - H block = 1 - 8).

1 - 2 - 3 - 4 - 5 - 6 - 7 - 8 - 9 - 10 - 11 - 12

Step difference between two floors = 20 steps

Step difference between two blocks = 10 steps

There are 6 errors in this code. Figure out the errors and write the corrected lines in the box below. One correction has been done for you. Do not entirely omit a code part, just correct the given code.

1	import java.util.scanner;
2	public class ErrorCorrection{
3	public static main(String[] args){
4	Scanner sc = new Scanner(System.in);
5	int steps = 0;
6	System.out.println("Which floor do you want to go?");
7	int floor to go = sc.nextInt();
8	System.out.println("Which block do you want to go?");
9	int block to go = sc.nextInt();
10	for(int i = 1; i <= floor to go; i++)
11	{
12	steps += 20;
13	}
14	for(int i = 1; i < block to go; i - -)
15	{
16	steps -= 10;
17	}
18	System.out.println("Steps covered:" , steps);
19	}
20	}

Write your corrections in the table given below. For better understanding, one error correction is shown.

Line Number	Fix
Line 3	public static void main(String [] args){
Line 16	steps += 10;
Line 1	import java.util.Scanner;
Line 10	for (int i=1; i < floor to go ; i++)
Line 14	for (int i=1; i < block to go ; i++)
Line 18	System.out.println("Steps covered:" + steps);

Question 4 [CO1] [4 Points]

[Answer on the question paper]

Illustrate the outputs of the following statements. Provide your workings on the answer script to verify your outputs. Your answer will not be accepted without the workings. All of the outputs must be in the question paper.

1	public class TracingB{
2	public static void main(String[] args){
3	int sum = 0, r = 0, i = 7, j = 6;
4	while (i < 100) {
5	r = (i%2) + 1;
6	if(r%2 == 0){
7	j = (++i) + j - r + 3;
8	continue;
9	}
10	sum = i % r + (i * r) - j + 38;
11	System.out.println(sum);
12	if (i > 12) {
13	sum += i/3 + r + 11;
14	break;
15	}
16	else{
17	i += 4;
18	}
19	}
20	System.out.println(sum);
21	}
22	}

Output
31 ✓
35 ✓
39 ✓
56 ✓